

Pako Maxence TEKOU

Applied Mathematics Student

Seeking a PhD position in Reinforcement learning / Optimization

/ Theoretical Machine Learning

Email: tekoumax15@gmail.com

Phone: +33 6 62 03 10 48

GitHub: github.com/maxencelebaron

LinkedIn: linkedin.com/in/pakomaxencetekou

EXPERIENCE

Machine Learning Research Intern, Inria (github.com/growingnet/gromo)

Paris-Saclay, France

Topics: Convex Optimization, Reinforcement Learning

April 2026 – Sept 2026

- Contributed to a team project deriving principled neuron addition criteria based on expressivity bottlenecks to determine when to grow a neural architecture versus continuing gradient-descent training
- Collaborated with a researcher to extend principled architecture growth criteria to mitigate plasticity loss in deep Reinforcement Learning policies
- Contributed to an open-source Python library and improved neural network training efficiency on GPU clusters

Data Science Intern, Crédit Mutuel Arkéa

Brest, France

Topics: Machine Learning, Clustering, Credit Risk Modeling

May 2025 – Aug 2025

- Segmented 2600 loan contracts using K-means clustering to surface distinct repayment behavioral patterns, enabling targeted recovery strategies
- Built decision tree and random forest classifiers to identify high recovery potential defaulted accounts
- Developed a probabilistic recovery scoring model that ranked defaulted loans by expected recovery amount

EDUCATION

École Normale Supérieure Paris-Saclay, M2 MVA (Mathematics, Vision, Learning)

Paris, France

Admitted via competitive selection from ENSAI's engineering programme

Sept 2025 – Sept 2026

- *Topics:* Convex Optimization, Signal Processing, Optimal Transport, Reinforcement Learning, Stochastic Calculus, Generative Modelling, Representation Learning, Time Series Analysis, Graphs in Machine Learning, Computational Statistics, LLM

École Nationale de la Statistique et de l'Analyse de l'Information (ENSAI), Cycle Ingénieur

Rennes, France

Bachelor's and Master's degree in Applied Mathematics and Computer Science

Sept 2023 – Sept 2025

- *Topics:* Statistical and Deep Learning, Natural Language Processing, Object-Oriented Programming, Algorithms and Data Structures, Stochastic Processes, Econometrics, Macroeconomics and Microeconomics

Lycée Albert Schweitzer, CPGE MPSI/MP*

Mulhouse, FRANCE

French Preparatory Classes for Elite Graduate Schools ('Grandes écoles')

Sept 2020 – Sept 2023

- *Topics:* Mathematics, Physics and Computer Science

RESEARCH ARTICLES & ACADEMIC PROJECTS

• **Should I train or should I grow? Single-Run Architecture Search by Network Growth**

S. Douka, A. Davey, S. Rivaud, T. Rudkiewicz, **P. M. Tekou**, et al. (Submitted to NeurIPS 2026, under review)

• **GrowRA: Growing and Allocating LoRA Rank with a Natural-Gradient Criterion**

T. Rudkiewicz, H. Mousset, E. Saad, A. Shilova, A. Davey, S. Rivaud, S. Douka, **P. M. Tekou**, et al. (Submitted to NeurIPS 2026, under review)

• **Optimization and Statistical Learning:** implemented gradient boosting algorithm from scratch and analyzed their convergence and regularization mechanisms from a functional optimization perspective

• **Reinforcement Learning:** implemented Q-Learning and Double DQN algorithms from scratch on MinAtar environments

• **Software Engineering:** built an interactive Rocket League platform with real-time database management used for performance tracking

LANGUAGES & SKILLS

• **Languages:** French (Native), English (Upper-Intermediate)

• **Soft Skills:** Collaboration on team projects, scientific communication, autonomous problem-solving

• **Programming:** Python (*PyTorch*, *NumPy*, *Pandas*, *Scikit-learn*), Java, R, SQL

• **Tools:** MLOps (*Weights & Biases/MLflow*, *Docker*, *Hydra*), Git, Excel, PowerPoint

• **Interests:** Basketball (National level in Togo), Afro Dance